

Practical

DevOps Practical - Git/GitHub, Docker, Jenkins(Installation & Pipeline)

1. I will be using Ubuntu. If you happen to use something different, chances are some commands might fail and it's completely fine. Take help from AI for that for correct command and way.
2. I will demonstrate the things below. For any doubts in the middle, please ensure to use AI and get it clear and if it is still confusing then feel free to reach out to me. I will even use AI in the middle of this practical.

- **Prerequisites Required On Your Device**

- Git installed
- Docker installed
- Java-21(for Jenkins installation. Lower version of java might fail installation)
- Github Account
- Dockerhub Account

- **Git, github(basic knowledge of below commands is a must for everyone)**

- Git commands - git init, git add, git commit, git push, git pull. No one knows everything about git/github but these commands are a must known for everyone.
- Clone the codebase - git clone
<https://github.com/pateldhruv2001/mandvi-devops-session-demo>
- Github Repo will contain many things which you can ignore.. If someone is interested then they can go through the other things as well.

- **Docker**

- Build image
 - `docker build -t devops-session-app:local .`
- Run image locally
 - `docker run -d --name devops-app-test -p 5001:5000 devops-session-app:local`
- Push to dockerhub
 - Token generation for pushing the image to dockerhub
 - `docker login -u <your-dockerhub-username>`
 - `docker tag devops-session-app:local <your-dockerhub-username>/devops-session-app:manual`
 - `docker push <your-dockerhub-username>/devops-session-app:manual`

- **Jenkins**

- Installation
 - `wget https://updates.jenkins.io/download/war/2.581/jenkins.war`
 - `nohup java -jar jenkins.war --httpPort=8081 &`
- Configure initial credentials
- Cidc pipeline jenkins job(pull code from github -> create docker image -> push to dockerhub -> run the container locally)

Rough Notes

BELOW CONTENT IS IN PARALLEL TO THE RECORDING. PLEASE WATCH THE VIDEO TO GET WHAT BELOW IS WRITTEN IN WHAT CONTEXT

This is what we will be doing today

Manually performing cicd using commands

Then will automate is by setting up jenkins and creating one pipeline within it to test.

Please go through the few important steps in above tab.

Lets begin by cloning the codebase.

Also learn basic linux commands - **cd, mkdir, ls, pwd**

I will be using these commands frequently.

Once git code based is cloned, use chatgpt or any AI tool and first rm the **.git** directory because it container my account's details(though not the password). I want you all to push this same codebase to your personal github account. Take help from chatgpt. I will move to next steps now.

You all might need to generate token for pushing the code to github. Let me show you quickly how to do that. Check all the checkboxes and generate token and the first time you run git push to push your code, and when asked for password, paste this as password. But before that, you all need to create one empty repository on github, Ask chatgpt that i have some code in local and need to push to github, it will show you the steps.

Our current codebase is in the app.py file which is in python and you can ignore that code. We are focussed on building the docker image right now. I have already created one docker file. It is simple. Just copied the code files into the image, installing the dependencies which is in the requirements.txt file and then running the container with the commands CMD has

Lets move forward

Make sure to run the docker build command from the location where your dockerfile is there.

The error says, we need to login to our docker first. Lets do that. Note that, docker will use the token of your account and not the password. So lets generate one quickly.

Now let me login to docker hub here by generating one token right now and using that.

Store the token securely somewhere and dont share with anyone!!

We have successfully build the docker image. Congrats!!

5001 is the port of our browser on which we will access the application. 5000 port is of the container port and we are saying that map 5000 port of container with my local system's 5001 port.

We have successfully spinned up the docker container using the docker image we recently built using the codebase we cloned from github!!

Or lets say now, we will feel the pain first. Let's say i make some code changes and want to depoy/run locally now again using docker ofcourse. The entire CICD commands we ran just now, need to be ran manually once again. CI -> Build the image and push to docker hub. CD -> use the same image and spin up the container. Simple.

LEt's make code changes and see the manual pain of running the commands. Code changes done, now re-run all the commands , build docker -> push to docker hub -> run the container. Giving each docker image a different tag which is here i have local-1.0, because in real world, we make sure, no docker image have the same tag for different codechanges because this will override our docker image. Already one container is running so we need to first stop that and the run this container, lets do that.

We had to run commands manually. Also, one this we missed, we need to push the images to the dockerhub also for storing there. People don;t store images locally, they always store on remote location which is dockerhu here. Local system can crash and we can loose the images. Lets push this current image to dockerhub.

Pushing images also has a specific format.

- docker tag <existing local image>:<tag> <your-dockerhub-username>/<existing image name>:<tag>
- docker push <your-dockerhub-username>/devops-session-app>manual

Once this is done, your image gets pushed to dockerhub. Pushing image to dockerhub is a part of CI stage.

Now, we saw all the manual commands we had to ran for a simple one line change. Create image, change the tag of the image, push to docker hub, stop the existing running container, run the container. This is manual pain. For 50 such github repositories, we can't afford this!! We need some automation tool and a script which does this!!

This is where jenkins comes into picture. We will install and setup jenkins and then will create one pipeline which does this automatically with just one click only.

What that pipeline will do, first we will push our code to githu(our recent code change is not on githu, let me revert that code first). Done. Once we make our code changes and then commit and push to github, the jenkins pipeline will fetch the latest code changes from github, login to docker hub, build the image, change the tag, push to docker hub, stop the existing running container and the will run the container again which is entire CICD pipeline done in just one click using jenkins.

Time stamp for jenkins - 34:00

Let's setup jenkins.

Wget - is used to fetch something from internet and we are providing the jenkins file we need to use.

Note that jenkins can be installed in many way, like docker-container, war file which uses java which we will do and also as a system process(systemctl).

Let's download the war file and then use that to setup the jenkins. We will use **nohup** which will create one file where we will see our jenkins logs.

Instead of running the wget command, you can directly click on the link which will also start the download process. Lets complete that and then will proceed. My internet is slow right now :)

nohup java -jar jenkins.war --httpPort=8081 &
8081 - is the port on which our jenkins will be running. Is on this port it shows some error like 'port already in use' in the nohup.out file which will be created right now, then please change the port to any other number like 9001, 9005, 8084 etc.

Nothing is running right now. Lets setup jenkins here. Might show some warning but if you see 'Jenkins is fully up and running' at the end then you are done. Congrats!! Jenkins is up.

Let's configure the credentials. Default password location will be showed, go to that file and copy the password and then lets see further.

"Welcome to Jenkins!" !! jenkins is all set. Note that for different use cases, people downloads plugins, like for sending email, there are plugins that need to be installed first. But for our usecase, we have what we need. Lets start by creating one pipeline.

You can ask chatgpt about what these configuration were. I have selected option "Execute Shell" means i need to execute some shell script. I have already prepare the script for you, you can directly copy paste and update few variable's values and use that.

It is present in the codebase only.

Update these variable's values- DOCKERHUB_USERNAME, DOCKERHUB_TOKEN, REPO_URL

Place your account specific values. Don;t use my url which contains my github username which is pateldhruv2001. This is the reason why initially i mentioned to push the codebase to your github account and use that url over here.

All set now, lets tirgger the pipeline. But wait!! How we will know on which port our application will be running?? It is in the script only. It is 5070. Again, while running the pipeline, if anyone gets error of port already in use that update this variable and re-run.

Let's first check if anything is running or not on this port. Nothing!

Now let's test our CICI automation pipeline. It will also push image to the dockerhub with this name,

```
IMAGE="$DOCKERHUB_USERNAME/devops-session-app"  
TAG="jenkins-native-${BUILD_NUMBER:-local}"
```

BUILD_NUMBER is the buil number which is which number of time we are build/runniing this pipeline. Let's test!! #1 is the build number.

Please go through the script and understand that well. That's important. Do not memorise the commands, but understand the flow, which can be seen in the logs also but script understanding is more important.

You shouldn't get this error because it is related to the local systems networking. I have recently made couple of changes to my local networking before recording this session. Let me quickly fix this. Also, note that I will also be using chatgpt but one should understand the response of the chatgpt well to be able to confidently and correctly use the commands,

Let me quickly fix that and let you know then we will proceed. But wait, our pipeline failed at the deploy stage. It has successfully build and pushed the image to docker hub, let's verify on docker hub. What is the name?? Devops-session-app:1

We verified, images successfully built and pushed to docker hub which implies CI pipeline is working fine. For CD, let me fix this issue first and then we will rerun the pipeline which will again create another image and then deploy that image.

I am using Claude Code instead of chatgpt because that's more useful for me and I have been using that more. Chatgpt I use for short and quick small requirements purpose.

People are using Claude Code for various purposes and at another level only, for example, let's say I have Elasticsearch/Grafana which is a monitoring tool, what I can do, I will generate one token from that with read only purpose, I will provide it to Claude and the URL of the monitoring tool and then inform Claude what to search for and provide me the details, I don't need to go to the portal manually and search things out, Claude will do it for me. Another example, I have made him find bugs in my code and test as well. For your surprise, the entire things in the codebase, including the ppt and documentation, all has been done through Claude only, I have informed what I need, what my intent and goal is, I reviewed it, I made couple of minor changes to it and tried to make the document as simple as possible. I know what I need to deliver, but for that, I need ppt, documentation and the sample codebase, I got it done from him. Note that students, one should be very well aware of the approach which Claude can't provide you. You only have to first figure out how to do and inform AI, since AI will do what you say and how you say!! Also, my main concern, this will cost money!!! So make sure, if you subscribe to this tool, you use it wisely and effectively!!

Okay, seems like our pipeline issue is resolved, Claude ran the command, the application should be live now on port 5070, let's see. No, it's just say what was the issue and where it was and told it is resolved, let's rerun the pipeline.

Recording this at 1:25 AM :) just for you guys!! Want you to understand these things and stand out in the interview!!

Also note that, during this entire process, many of you might face issues, but make sure to use chatgpt, take help from instructor wherever possible, since this once you will understand, you will already be ahead of the majority students at your level.

Big Congratulations to all of you!! YOU have just built an end to end CICD pipeline on jenkins.!!!

Now lets make one simple code change and commit it and push to github and rerun the pipeline to see the changes. Note that how fast now the process is in place after code change is done, just one click!! Lets see. Just make sure you push your code to github before running the pipeline since thats how our pipeline works. Let me show you exactly where it is configured in jenkins pipeline script. If you don't get then please learn the git clone and git pull commands. Thats imp.

Thats it guys!!!! One click and all manual efforts went and our CICD is doing its work great!!! It is working as expected. Try by yourself by making some code changes and running the pipeline and understsing it better how it works!!!

LEt me revert the cpde and again test the pipeline. Notice, how fast things are now compared to before where we ran manual commands for just this single codebase. Now if we have 50 github repositories, we will just create the duplicates of this pipeline we just created and just update the github repo url.

Also for this run, note that we have ran pipeline 3 times, and 3 docker images would be created and are pushed to github, lets see that first. Its there, verified!! Lets revert the codebade and run the pipeline after pushing to github. Everything is working as expected.

Thats all for this session guys.

A few things is in the codebase which is githubaction which is this. This also used for CICD and running the same things we ran on jenkins. In jenkins here, CI and CD was running on our localsyste. In github actions, CI I have ran on github side and CD on my local. The complete documentation of this will be found here in the documentation.

Also, there are couple of things more in the documentation like terraform, scaling, architecture, you can COMPLETELY IGNORE THAT!! I just wanted to showcase to the students from my side during the session time! If and only if someone is more and more curious about that then can get an overview of it but no need to perform practical and also its fine if you don't go there since that would become much more complicated. So please ignore that!!

Now the last clean up process.

Note that, i will delete my docker hub token since it is clearly visible in the recording. Now it is invalid, We will stop the jenkins process now. Run **ps aux | grep jenkins** basically list all processes and grep those which contains the word **jenkins**. **kill -9 <processid>** is basically to kill the process and jenckns should not be running now. Its sopped and just the cache issue. Verify the process should not running. Now lets stop the running containers since they are consuming cpu. Ignore the one which i have, they won't be there on your system. If multiple containers are there then make sure to delete the one which we were using. Great. Now lets remove the jenkins config file which jenkins creates when it was setup which is at location here **~/jenkins** within this all the configurations/data are still there, we have just

stopped the process and not the data. Next time when you setup the jenkins using the steps we performed, it will use this same location and it will already show the jenkins pipeline we created and 4 jobs we ran. To completely remove the data also then run **rm -rf ~/.jenkins**

Its gone!!

Its done now, i really hope all of you have understood what we did today and i understand some of you might have face issues, please check with chatgpt and the lab instructors to get that resolved. Even if this will take 1-2week full dedication or even 1 month or more then also this is completely justified for you guys. So please make sure you understand this will and do some changes and test it out how it works in details.

If anyone faces are issues and is not getting resolved by chatgpt or any other AI tool, please reach out to the HoD with all your query and I request the HoD to forward them to me and I would be happy to check on that and get back to you asap.

Thanks All for watching this till the end.

Thanks all!!!